Vasileios Gkolemis

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Education

Harokopio University of Athens — Athens, Greece
PhD candindate in Explainable Machine Learning (XAI) — *Mar. 2022 –*
Supervisors: Prof. Christos Diou, Prof. Eirini Ntoutsi, Theodore Dalamagas

University of Edinburgh — Edinburgh, Scotland, UK
MSc Operational Research with Data Science - with Distinction 78/100 — *Sep. 2019 – Oct. 2020*
Dissertation: Robust Optimisation Monte Carlo for Likelihood-Free Inference

University of the Basque Country — Vitoria-Gasteiz, Spain
Erasmus in "Escuela Universitaria de Ingeneria" — *Jan. 2016 – Jun. 2016*

Aristotle University of Thessaloniki — Thessaloniki, Greece
Diploma in Electrical and Computer Engineering - 7.56/10 — *Sep. 2011 – Oct. 2017*
Dissertation: Stereoscopic vision using artificial neural networks

High School — Athens, Greece
The American College of Greece — *Sep. 2008 – Jun. 2011*
Final GPA: 18.4/20

Professional/Research Experience

Research Assistant at ATHENA Research and Innovation Center — Athens, Greece
Research Projects: XMANAI Explainable AI — *Nov. 2020 –*

Research Assistant at Multimedia Understanding Group — Thessaloniki, Greece
Research Projects: BigO, i-prognosis — *Jan. 2018 – Aug. 2019*

Workshops - Online courses

Summer Schools

- **PAISS 2021**: Artificial Intelligence Summer School
- **Fairness in AI**: First Winter School on Fairness in AI

Coursera Online Courses

- **Deep Learning Specialization**: (i) Structuring Machine Learning Projects, (ii) Improving Deep Neural Networks: Hyperparameter tuning, Regularization and Optimization, (iii) Neural Networks and Deep Learning
- **TensorFlow Developer Professional Certificate**: (i) Introduction to TensorFlow for Artificial Intelligence, Machine Learning, and Deep Learning
- **Machine Learning Engineering for Production**: (i) Introduction to Machine Learning in production

Knowledge - Technical Skills

Backgrounf knowledge:

- **Machine Learning**: Bayesian Modeling, Likelihood-free Inference, Probabilistic Graphical Models, Gaussian Processes
- **Deep Learning**: CNN (expertise in Computer Vision applications), Graph Deep Learning
- **Optimization**: Linear (Simplex, Interior Points), Non-Linear (First/Second Order Methods), Bayesian Optimization

Machine Learning Skills:

- **Deep Learning**: PyTorch, Tensorflow, Keras, JAX
- **Probabilistic Modeling**: Pyro, GPyTorch, PyMC3, ELFI
- **Data Science**: Hadoop, Spark
- **Scientific Computing Libraries**: Numpy, Scipy, Pandas, MatplotLib, openCV, Scikit-learn

IT Skills:

- **Programming languages**: Python, C, Java, R, Matlab, Scala
- **DevOps**: Docker
- **Parallel Programming**: CUDA, Pthreads
- **Database Development**: MongoDB, Cassandra, MySQL
- **Operating Systems**: Linux
- **Other skills**: Git, Github, LaTex

Foreign Languages

- **English**: Proficient User • C1 • IELTS: 7.5/10 Band Score
- **Spanish**: Independent User • B2 • Delle
- **French**: Basic User • A2 • Delf

About me

I am a PhD candidate in Explainable AI (XAI) with uncertainty quantification under the supervision of Prof. Christos Diou, Prof. Eirini Ntoutsi and Prof. Theodore Dalamagas. My PhD research focuses on designing explainability techniques for probabilistic ML models, like Bayesian Neural Networks. I believe that uncertainty and explainability are the two fundamental components for Trustworthy ML.
Before that, I completed my diploma in electrical and computer engineering at the Aristotle University of Thessaloniki in November 2017. Throughout my studies, I mainly focused on machine learning with applications in computer vision. In my dissertation, I was primarily involved in the domain of Deep Learning. Afterwards, I worked as a research associate in the EU-funded project BigO. At September 2020, I completed my MSc in Operational Research with Data Science at the University of Edinburgh, where I worked on Probabilistic (Bayesian) Machine Learning and, especially, in Likelihood-free inference methods. As part of my dissertation, I implemented the ROMC method - which I still maintain - at the software package ELFI. Apart from the world of engineering, I love traveling, playing football and watching movies.